

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 2



Books

Text Book

Harry Lewis, Christos Papadimitriou.

Elements of the Theory of Computation.

Second Edition, Pearson Education, Asia. 1998

Reference Book

John C Martin

Introduction to Languages & The Theory Of Computation.

Third Edition. Tata McGraw Hill

How to do well...

1. you must learn the concepts well;
2. if you do learn the concepts, there is very little else (facts, etc.) to learn; you can do really well!
3. You must do problems. There's no replacement for this.
4. Attending lectures is highly advised!
5. It will be very hard to learn the concepts by yourself or from textbook. Topics get quickly hard.
6. Come regularly to tutorials; you will learn a lot by working out problems and learn from fellow students

Mathematical Preparation

Set

Subset

Operations on Sets – Union, Intersection etc.

Power Set - 2^A or $P(A)$

Let A be a set & Π be subset of 2^A

Π is a *partition* of A if

1. $\emptyset \notin \Pi$
2. Distinct members of Π are disjoint
3. $\cup \Pi = A$

Result - For any finite set A , $|2^A| = 2^{|A|}$

Proof by *Mathematical Induction*

In the course, we are going to use Mathematical Induction many times

Mathematical Preparation

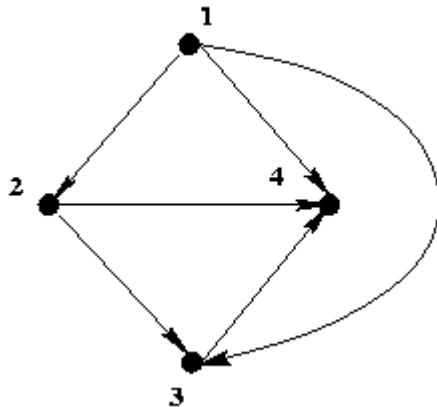
Relations

Binary Relation R on A & B is a subset of $A \times B$

Binary Relation R on A is a subset of $A \times A$

Binary Relation R on A can be represented as *directed graph*.

Arrow is drawn from a to b if and only if $(a, b) \in R$



Mathematical Preparation

Special Type of Relations on set A

1. Reflexive
2. Symmetric
3. Transitive
4. Antisymmetric

Relation R which satisfies $1 \wedge 2 \wedge 3$ is called an **equivalence relation**

Relation R which satisfies $1 \wedge 3 \wedge 4$ is called **an partial order**

A partial order in which any two elements are comparable is called an **Total Order**

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Equivalence Classes

Given an equivalence relation R on a set A , the equivalence class of an element ' a ' (denoted as $[a]$) is the set of all elements in A that are related to ' a ' by relation R .

Theorem:

Let R be an equivalence relation on a nonempty set A . Then the equivalence classes of R constitutes a partition of A

Mathematical Preparation

A **path** in a binary relation R from element a to element b is a sequence of elements $(a_1, a_2, \dots, a_n)$ where $a_1 = a$, $a_n = b$, and for each i from 1 to $n-1$, the pair $(a_i, a_{i+1}) \in R$

The **length** of the path $(a_1, a_2, \dots, a_n)$ is **n** .

The path $(a_1, a_2, \dots, a_n)$ is a **cycle** if all a_i 's are distinct and also $(a_n, a_1) \in R$

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Theorem:

Let R be a binary relation on a finite set A & let $a, b \in A$. If there is a path from a to b in A , then there is a path of length at most $|A|$.

Proof:

Let $(a_1, a_2, \dots, a_n)$ where $a_1 = a$, $a_n = b$, be a path of shortest length & suppose $n > |A|$.

By **Pigeonhole Principle**, $\exists$ an element of A that repeats on the path, say $a_i = a_j$ for some $1 \leq i < j \leq n$.

Then $(a_1, a_2, \dots, a_i, a_{j+1}, \dots, a_n)$ is a shorter path from a to b .

Contradiction to the assumption that $(a_1, a_2, \dots, a_n)$ is the shortest path from a to b .

Mathematical Preparation

Finite & Infinite Sets

We call two sets A & B **equinumerous** if there is a bijection $f : A \rightarrow B$

A set is **finite** if it is equinumerous with $\{1, 2, \dots, n\}$ for some natural number n .

A set is **infinite** if it is not finite.

A set is **countably infinite** if it is equinumerous with $\mathbb{N}$ the set of natural numbers.

Countably infinite set can be **enumerated** as $\{a_0, a_1, a_2, \dots, a_n, \dots\}$

A set **is countable** if it is finite or countably infinite

A set that is not countable is **uncountable**